Tao Jing (荆韬)

Beijing 100084, China · Tsinghua University · jingt20@mails.tsinghua.edu.cn
astro-jingtao.github.io · astro-jingtao · 0009-0004-6271-4321

Research Interests

- Investigate stellar feedback and its impact on star formation and quenching processes at sub-galactic scales using multi-band imaging and integral field spectroscopy.
- Study the interactions within the multiphase interstellar medium.
- Explore the microstructure of shock- or photon-ionized nebulae in the Milky Way and nearby galaxies to refine ionization models and improve hydrodynamic simulations.
- Develop and tailor advanced machine learning techniques for efficient analysis and interpretation of astronomical datasets.

Education

- **Doctor of Philosophy**, Astronomy
  Department of Astronomy, Tsinghua University (THU) — 2020 - Now
- **Bachelor of Science**, Astronomy
  Department of Astronomy, Xiamen University (XMU) — 2016 - 2020

Publications

- **Jing, T.** and Li, C. (2025), Correlations of ALMA CO(2-1) with JWST mid-infrared fluxes down to scale of $\lesssim$ 100 parsec in nearby star-forming galaxies from PHANGS. arXiv e-prints, arXiv:2511.10464
- **Jing, T.** and Li, C. (2025), Regression for Astronomical Data with Realistic Distributions, Errors, and Non-linearity. AJ, 170, 45, 10.3847/1538-3881/add891
- **Jing, T.** and Li, C. (2024), On the Origin of Quenched but Gas-rich Regions at Kiloparsec Scales in Nearby Galaxies. ApJ, 975, 17, 10.3847/1538-4357/ad7367
- **Jing, T.**, Li, C., Yan, R., Cheng, C., Zhang, W., Ji, X., Li, N., Wang, J., Wu, C., and Yuan, H. (2024), Criss-cross Nebula: A Case Study of Shock Regions with Resolved Microstructures at Scales of 1000 au. ApJ, 962, 51, 10.3847/1538-4357/ad11d1
- Sui, C., Mao, Y., Zhao, X., **Jing, T.**, and Wandelt, B. D. (2025), How to evaluate the sufficiency and complementarity of summary statistics for cosmic fields: an information-theoretic perspective. arXiv e-prints, arXiv:2511.08716
- Ma, C., Sun, Z., **Jing, T.**, Cai, Z., Ting, Y.-S., Huang, S., and Li, M. (2025), Can AI Dream of Unseen Galaxies? Conditional Diffusion Model for Galaxy Morphology Augmentation. arXiv e-prints, arXiv:2506.16233, 10.48550/arXiv.2506.16233
- Guo, R., Li, C., Zhou, S., Li, N., **Jing, T.**, and Cheng, Z. (2025), Mapping Dust Attenuation at Kiloparsec Scales. II. Attenuation Curves from Near-ultraviolet to Near-infrared. Research in Astronomy and Astrophysics, 25, 065017, 10.1088/1674-4527/add673
- Sui, C., Zhao, X., **Jing, T.**, and Mao, Y. (2023), Evaluating Summary Statistics with Mutual Information for Cosmological Inference. Machine Learning for Astrophysics, 30, 10.48550/arXiv.2307.04994

Collaborations

- Affordable Multiple Aperture Spectroscopy Explorer (AMASE) project: a in-processing project that will pair 100 identical multi-fiber spectrographs to map the ionized gas in the Milky Way and nearby galaxies at the spatial resolution of half an arcminute and a spectral resolution of R=15,000.

  I am in Science Team of AMASE and work for the HII region sample selection and construct mock data cubes from simulations. I am also leading a subproject to study microstructures in shocked regions.

Approved Proposals

- PMO Delingha Millimeter Telescope 2022 (30.5 hours awarded) — PI
  Title: Detecting ^{12}CO in the Criss-cross Nebula to study shock-molecular cloud collision.
- SITELLE@CFHT 2022B (5.5 hours awarded) — PI
  Title: High resolution IFS on Criss-cross Nebula as a case study of microstructure of shock regions
- MeerKAT 2022 (7.5 hours awarded) — PI
  Title: High resolution HI observation on Criss-cross Nebula: the case study for shock-cloud interaction
- SITELLE@CFHT 2024B (PI: Xiao Li, 5.6 hours awarded) — CoI
  Title: Large FoV IFS on UGC1382 as a case study of quenched but HI-rich galaxies

Technical Skills

- **Programming Language**: Python, C, FORTRAN, JavaScript, Mathematica, LaTeX
- **Ionization Model**: MAPPINGS V
- **Spectrum Analysis**: pPXF, self-developed code
- **Simulation Post-processing**: COLT, CHIMES
- **Machine Learning Packages**: TensorFlow, JAX, PyTorch, scikit-learn, AutoGluon

Selected Talks and Posters

- The 12th Sino-German Meeting on Galaxy Formation — 2025
  Poster + Flash talk, Chengdu, China
- Decoding Galactic Evolution through the Interplay of the Multi-Phase Interstellar Medium — 2025
  Poster + Flash talk, Nagoya, Japan
- The 12th Symposium on Interstellar Physics and Chemistry — 2025
  Oral talk, Zhangye, China
- Symposium on Evolutionary Laws of Galactic Ecosystems over Billions of Years — 2025
  Poster + Flash talk, Beijing, China
- The 27th Guoshoujing Symposium on Galaxies and Cosmology — 2025
  Oral talk, Changsha, China
- AMASE Science Meeting — 2025
  Oral talk, Xiangtan, China
- JWST Discussion Workshop — 2025
  Invited talk, Given remotely
- The Ecosystem of Gas and Dust in Galaxies Near and Far — 2024
  Oral talk, Chiang Mai, Thailand
- FAST Deep Sky Survey Annual Meeting 2024 — 2024
  Oral talk, Beijing, China
- THU Galaxy & Cosmology Seminar — 2024
  Invited talk, Beijing, China
- The 11th Symposium on Interstellar Physics and Chemistry — 2024
  Oral talk, Nanjing, China
- AMASE Science Meeting — 2024
  Two oral talks, Beijing, China
- The 26th Guoshoujing Symposium on Galaxies and Cosmology — 2024
  Oral talk and Poster, Suzhou, China
- Resolving Galaxy Ecosystems Across All Scales 2023 — 2023
  Oral talk, Hongkong, China

- ACAMAR Gas in Galaxies Workshop 2023 — 2023
  Oral talk and Poster, Perth, Australia
- LAMOST MRS-N and Star Formation Evolution Seminar — 2023
  Oral talk, Xuyi, China
- SDSS Collaboration Meeting 2021 — 2021
  Oral talk, Given remotely
- The 23th Guoshoujing Symposium on Galaxies and Cosmology — 2021
  Oral talk, Hangzhou, China

Teaching, Professional Services, and Student Supervision

- Qianyi Liu (刘谦益), undergrad at THU — 2025 - now
- LOC, Symposium on Evolutionary Laws of Galactic Ecosystems over Billions of Years — 2025.06
- LOC, AMASE Science Meeting — 2024.05
- Organizing Committee of THU Data Science Club/Machine Learning Session — 2021 - 2023
- Teaching Assistant, Observational Cosmology: From the Solar System to the Depths of the Universe — 2020 - 2021

Honors and Awards

- Tsinghua University Comprehensive Excellence Scholarship (First Prize) — 2024 - 2025
- Tsinghua University Comprehensive Excellence Scholarship (Second Prize) — 2023 - 2024
- AMD Scholarship — 2020 - 2021
- Shanghai Astronomical Observatory Guangqi Scholarship (First Prize) — 2018 - 2019
- Xiamen University Tan Chong International Limited Scholarship — 2017 - 2018

References

- Prof. Cheng Li — Tsinghua University
  cheng.leech@gmail.com
- Prof. Jing Wang — Peking University
  jwang_astro@pku.edu.cn
- Prof. Renbin Yan — The Chinese University of Hong Kong
  rbyan@cuhk.edu.hk